

Pion Production and Symmetry Energy in Intermediate-Energy Heavy-Ion Collisions

Background

A central question in nuclear physics is how nuclear matter behaves at densities above saturation. Heavy-ion collisions at intermediate energies provide one of the few ways to access this regime in the laboratory.

Among different observables, pions are often considered particularly useful, since they are mainly produced during the high-density stage of the reaction. In this context, quantities such as the π^-/π^+ ratio and collective flow have been discussed as possible probes of the symmetry energy.

What I worked on

In my recent work, I used the LQMD transport model to study pion production in heavy-ion collisions. The main focus was on understanding how microscopic ingredients of the model influence the final observables.

In particular, I looked at:

- pion production through Δ resonance excitation,
- in-medium effects such as modifications of Δ properties and pion–nucleon interactions,
- the sensitivity of π^-/π^+ ratios and flow observables to the symmetry energy.

Rather than only comparing results, I tried to understand where the sensitivities come from in the reaction dynamics.

Results

The calculations show that pion observables are noticeably affected by in-medium effects, especially through changes in Δ dynamics and pion propagation.

Both the π^-/π^+ ratio and collective flow show sensitivity to the density dependence of the symmetry energy, although this sensitivity depends on how in-medium effects are treated.

From a systematic comparison within the model, I obtained a constraint on the slope parameter of the symmetry energy, $L \approx 42 \pm 25 \text{ MeV}$, with uncertainties mainly coming from model assumptions.

What I would like to do next

For my PhD, I would like to continue working on transport descriptions of heavy-ion collisions, with an emphasis on understanding the underlying physics rather than focusing on a specific model implementation.

I am particularly interested in how different transport approaches treat in-medium effects, resonance dynamics, and particle production, and how these differences influence observable signals related to the symmetry energy.

At the same time, I am open to working with different transport frameworks and to adapting my background to the specific methods used in a given group.

More generally, I would like to better understand what can be robustly extracted about dense nuclear matter from heavy-ion collisions, especially in connection with current and future experiments.